

Verification of a Local VQE Stack for a Thio-Michael Addition

Twenty exact checks, an ansatz comparison, and the
qubit count at which a laptop stops being enough

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Abstract

This report documents the verification of a Qiskit Aer VQE workflow built for the C–S bond-forming region of a thio-Michael addition — acrylamide plus methanethiolate — following the simulator protocol of CovAngelo (arXiv:2604.10487). It is a **verification report, not a chemistry report**: the molecular Hamiltonians for the two target geometries have not been built, for a stated and specific reason, and §6 says exactly what remains and what it would cost.

What is established is the correctness and the cost boundary of the solver. Twenty checks against exact classical answers pass on a synthetic but structurally real CAS(4e,4o) active space constructed through the same OpenFermion path the real Hamiltonian would use. On that system UCCSD recovers 99.74% of a 374.58 mHa correlation energy, landing 0.978 mHa above exact sector FCI, with particle-number leakage of 2.8×10^{-16} and no penalty term.

Four findings changed the workflow’s defaults and are reported because each overturned an assumption. Spin-orbital ordering between OpenFermion and qiskit-nature differs by a permutation that is a similarity transform, so it preserves the spectrum exactly while corrupting every determinant energy — no eigenvalue check can detect it, and it is therefore verified determinant by determinant. Parameter-shift gradients are invalid for UCCSD and the runtime refuses them rather than returning a wrong gradient silently. Qiskit Aer terminates the process on Windows for circuits deeper than roughly 10^4 gates, tracking depth rather than qubit count, and the workflow executes every Aer call on an enlarged stack as a result. And k -UpCCGSD at $k \geq 2$ is measurably more accurate than UCCSD on this system, while ADAPT-VQE selected all 26 of 26 pool operators — rediscovering UCCSD exactly, for the same energy at 7.3 times the cost.

Measured scaling places the practical ceiling for this workflow at 16 qubits on a laptop; 20 qubits is projected at 295 hours for a 50-iteration run.

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1 Introduction

1.1 The target system

A thio-Michael addition is the covalent-inhibitor chemistry that motivates the CovAngelo study: a thiolate nucleophile attacks the β -carbon of an α, β -unsaturated carbonyl, forming a C–S bond. The quantity of interest is the energy difference between a transition-state-like geometry and separated reactants, in an active space small enough for a quantum treatment.

Table 1: The intended calculation, and its relationship to the published study.

	CovAngelo	This workflow
Ansatz	UCCSD (reported “most reliable”)	UCCSD
Orbitals	MP2 natural	MP2 natural
Solvation	PCM, $\varepsilon = 4$	C-PCM, $\varepsilon = 4$
Geometries	TS and separated reactants	ts and far
Basis	6-31++G**, aug-cc-pVDZ	aug-cc-pVDZ
Reference	FCI	CASCI = FCI in the space
Hamiltonian	ECC-DMET, optimised bath	CAS(ne,no)
Hardware	IQM Garnet, 8 qubits	not attempted

Scope of comparability

The Hamiltonian row is the one that matters. CovAngelo’s ECC-DMET embedding with quantum-information-optimised bath orbitals is the paper’s actual novelty, is not available in PySCF or Qiskit, and reconstructing it is a research project in its own right. Absolute energies from this workflow are **not comparable to that paper’s Table 2**. What carries over is the methodology and the VQE-against-exact-diagonalisation agreement.

Two further observations from the published work shaped this one. Their hardware run missed FCI by 2.058–2.943 Ha — roughly 1,300 to 1,850 kcal/mol — which the paper attributes to device error rates; their simulator results are the reproducible ones, and they used Qiskit AerSimulator to produce them. And their own numbers show orbital choice moving the VQE energy by 0.887 Ha, which is far larger than any ansatz or optimiser effect they report. That is why MP2 natural orbitals are this workflow’s default rather than an option.

1.2 What this report establishes, and what it does not

1. **Established:** the solver is correct against exact classical answers on a structurally real active space, and its cost is characterised as a function of qubit count and ansatz.
2. **Established:** four implementation findings, each measured, each of which would have produced a wrong or unobtainable answer if left unaddressed.
3. **Not established:** any energy for acrylamide plus methanethiolate. No Hamiltonian for either geometry has been built. §6 states why and what is required.

2 Verification protocol

2.1 The synthetic system

`selftest` constructs a random but *structurally real* CAS(4e,4o) Hamiltonian — correct operator structure, correct symmetries, correct sparsity pattern — through the same OpenFermion path the molecular Hamiltonian would take, and then checks every component of the stack against exact classical answers computed independently. It requires neither PySCF nor a cache, which is what allows the solver to be validated on a machine where the chemistry cannot be built.

Property of the verification system	Value
Qubits	8
Pauli terms	361
Reference determinant	−23.6854521491 Ha
Exact sector FCI	−24.0600287933 Ha
Correlation energy	−374.577 mHa
UCCSD parameters	26
UCCSD depth / two-qubit gates	1,646 / 1,376

2.2 Results of the checks

All twenty checks pass. The run reproduced for this report took 168.5 seconds.

2.3 The check that matters most

OpenFermion emits spin orbitals **interleaved**: orbital $2p$ is α_p , orbital $2p + 1$ is β_p . The `JordanWignerMapper` in `qiskit-nature` expects them **blocked**: all α , then all β . At eight qubits the

Table 2: Verification checks and their measured residuals. Grouped by what each one pins down.

Check	Residual	What it excludes
<i>Operator construction</i>		
Determinant energy formula	-2.49×10^{-14} Ha	classical/qubit mismatch
Operator conversion, spectra	7.11×10^{-14} Ha	wrong Hamiltonian
Interleaved ordering, per det.	7.11×10^{-15} Ha	qubit relabelling
Blocked ordering, per det.	7.11×10^{-15} Ha	qubit relabelling
Blocked map is non-trivial	[0, 4, 1, 5, 2, 6, 3, 7]	a vacuous test
<i>Ansatz</i>		
$\theta = 0$ is the reference det.	-1.43×10^{-12} Ha	wrong initial state
$\theta = 0$ electron number	$\langle \hat{N} \rangle = 4.0000000000$	wrong filling
UCCSD conserves N	leak 6.80×10^{-16}	symmetry breaking
<i>Simulation and gradients</i>		
MPS agrees with statevector	5.36×10^{-10} Ha	truncation error
Parameter shift correctly refused	—	silent wrong gradient
Finite differences step-stable	1.11×10^{-8}	step-size artefacts
Engines expose same parameters	aer 26, numpy 26	divergent code paths
NumPy engine reproduces state	2.91×10^{-14}	engine disagreement
NumPy engine reproduces energy	1.36×10^{-12} Ha	engine disagreement
Adjoint matches finite diff.	6.74×10^{-9}	gradient error
<i>Converged state</i>		
Variational bound respected	+0.9782 mHa	below-reference results
Improves on reference det.	99.74% recovered	a non-functioning VQE
Still in the target sector	leak 2.78×10^{-16}	sector escape
Converged state is a singlet	$\langle \hat{S}^2 \rangle = 1.12 \times 10^{-4}$	spin contamination
Optimiser made real progress	14 iterations	a frozen start

permutation is [0, 4, 1, 5, 2, 6, 3, 7].

Why this cannot be caught by an eigenvalue test

A bit permutation is a similarity transform. Apply the wrong one and the resulting Hamiltonian has an **identical spectrum** to the correct one — same ground-state energy, same gap, same everything a spectral check inspects — while the energy of every individual determinant is wrong. A VQE on that operator converges cleanly to a number that is not the molecule’s energy, and no check on eigenvalues can distinguish the two.

The ordering is therefore verified *determinant by determinant*, in both conventions, over all eight determinants of the verification system, and the permutation is separately asserted to be non-trivial so that the test cannot pass vacuously on an identity map.

3 Ansatz comparison

Table 3: Five ansatz configurations on the 8-qubit verification system, scored against exact sector FCI. Correlation energy available: 374.58 mHa.

Ansatz	Params	Depth	Error (mHa)	Correlation	Time (s)
UCCSD	26	1,646	+0.978	99.739%	8.7
k -UpCCGSD, $k=1$	14	544	+7.939	97.881%	6.1
k -UpCCGSD, $k=2$	28	1,086	+0.342	99.909%	95.3
k -UpCCGSD, $k=3$	42	1,628	+0.136	99.964%	228.4
ADAPT-VQE	26	1,590	+0.983	99.738%	63.9

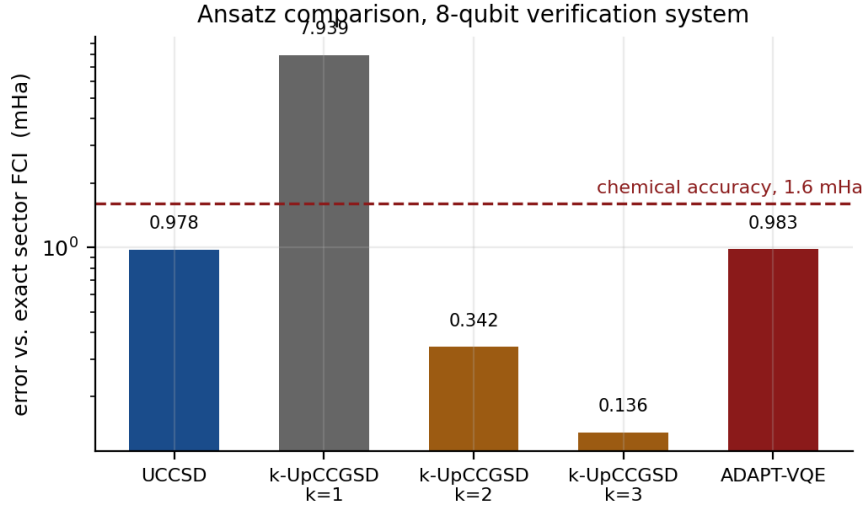


Figure 1: Error against exact sector FCI, logarithmic. Every configuration except k -UpCCGSD at $k=1$ clears chemical accuracy on this system.

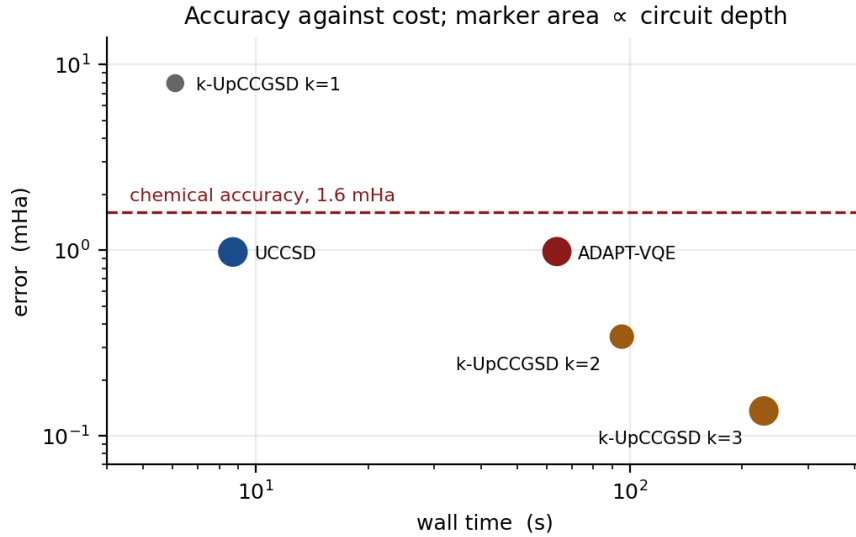


Figure 2: The same five runs as accuracy against wall time; marker area is proportional to circuit depth. UCCSD occupies the corner: within a factor of three of the best accuracy, at one twenty-sixth of the cost.

Two results are worth separating.

k -UpCCGSD at $k \geq 2$ is more accurate than UCCSD, by a factor of 2.9 at $k=2$ and 7.2 at $k=3$. The reason is structural: its excitations are *generalized*, not restricted to occupied $\rightarrow$ virtual, so it reaches correlation that singles-and-doubles from the reference determinant cannot. The cost is not proportional — $k=3$ is 26 times slower than UCCSD for a $7.2\times$ accuracy gain — so UCCSD remains the default at this size and k -UpCCGSD becomes the choice above ten qubits, where UCCSD’s $O(N^4)$ parameter count stops being affordable.

ADAPT-VQE selected all 26 of 26 pool operators. It rediscovered UCCSD exactly, arriving at the same energy to within 0.005 mHa, having spent 7.3 times the wall time to do so. This is not a failure of ADAPT; it is a statement about when ADAPT is worth running. The method earns its cost when the pool is far larger than the number of operators actually required, so that a gradient-driven search can skip most of it. At eight qubits with a singles-and-doubles pool, it is not, and the search is pure overhead. ADAPT is implemented and available, and `-ansatz auto` does not select it.

4 Cost boundary

Table 4: Measured UCCSD cost on 8 CPU cores, statevector, with a molecular-scale observable. “50 iterations” is the projected wall time of a converged VQE.

Register	Params	Depth	1 energy	1 gradient	50 iterations
8q (4e,4o)	26	1,646	0.77 s	0.7 min	0.6 h
12q (6e,6o)	117	11,127	0.54 s	2.1 min	1.8 h
14q (7e,7o)	204	22,163	2.38 s	16.2 min	13.5 h
16q (8e,8o)	—	—	not reached in ~ 20 min		

Table 5: Measured k -UpCCGSD ($k=2$) cost, same machine, $O(N^4)$ Pauli count, 50-iteration VQE.

Register	Params	Depth	1 energy	1 gradient	50 iterations
12q	66	3,226	0.23 s	31 s	0.43 h
14q	90	4,908	0.69 s	124 s	1.7 h
16q	120	7,040	1.91 s	459 s	6.4 h
18q	152	9,780	9.33 s	2,846 s	40 h
20q	190	13,054	55.8 s	21,271 s	295 h

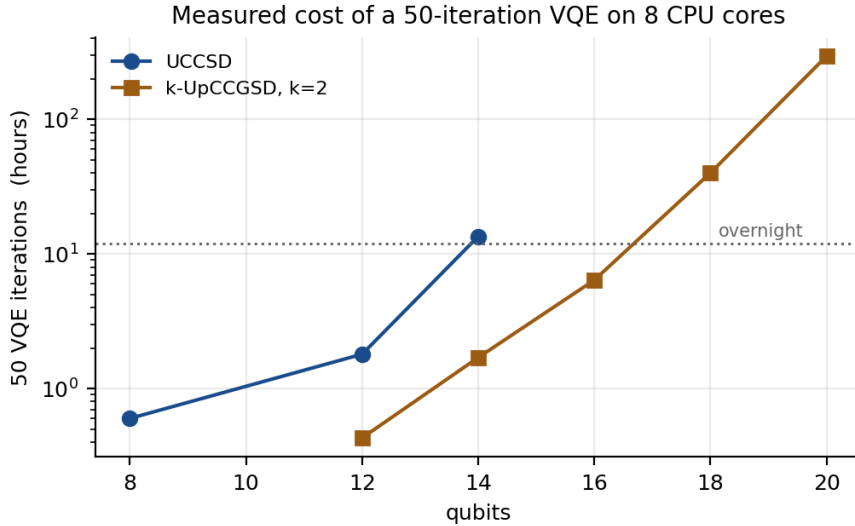


Figure 3: Projected wall time for a 50-iteration VQE against register size, for both ansatz families. The vertical axis is logarithmic; the curves are steeper than exponential in qubit count because parameter count and depth grow simultaneously.

The boundary

The wall is not memory. A 20-qubit statevector is 16 MB. The cost scales as (parameters $\times$ depth $\times 2^n \times$ terms), and on this workflow every one of those four factors grows with register size at once. **16 qubits is the practical ceiling on a laptop; 20 is out of reach.**

For UCCSD specifically, depth roughly doubles per two qubits while parameter count grows as $O(N^4)$, so gradient cost compounds in both directions: 8q and 12q are comfortable, 14q is an overnight run, and 16q was not reached to completion.

5 Implementation findings

Each of the following was measured, and each overturned the assumption the workflow was originally written under.

5.1 Parameter binding is worth $31.5\times$

Binding parameters with `assign_parameters` costs 419 ms per circuit, because each of the 26 UCCSD parameters appears in roughly eight gates as a symbolic expression evaluated in Python. The Aer execution itself costs 66 ms. Handing Aer the unbound circuit plus a bind table moves the work into C++:

53-vector gradient batch	build	run	total
<code>assign_parameters</code>	33.55 s	9.18 s	42.73 s
<code>parameter_binds</code>	0.04 s	1.32 s	1.36 s

Results agree to 3.9×10^{-14} . End to end this took the self-test VQE from 407 s to 47 s.

5.2 Parallelism is method-dependent, and the obvious answer is wrong

	threads = 1, experiments = 8	threads = 8, experiments = 1
statevector	70.8 ms/circuit	36.0 ms
MPS	83.9 ms	153.9 ms

A shallow circuit is too small to thread, so circuit-level parallelism wins; a deep one threads well. UCCSD is deep. Defaults are set per method accordingly.

5.3 Statevector beats MPS here, as predicted

UCCSD’s 1,376 entangling gates saturate the bond dimension — at eight qubits $\chi \leq 2^{n/2} = 16$ — so MPS compresses nothing and pays pure overhead, losing by $2.3\times$. **The MPS memory knob is ansatz depth, not qubit count.** MPS remains the method that scales for weakly entangled circuits at larger n , and the converged energy is cross-checked against exact statevector either way.

5.4 Parameter shift is invalid for UCCSD

Each UCCSD parameter drives roughly eight rotations, so the two-term shift rule silently returns a wrong gradient. `parameter_shift_is_exact()` detects the condition and refuses. The textbook alternative — qiskit-algorithms’ adjoint `ReverseEstimatorGradient` — is numerically correct, agreeing with finite differences to 1.7×10^{-11} , but took 619 s for a single 26-component gradient against 1.36 s for batched finite differences. It is not used, and `requirements.txt` documents why it is commented out rather than absent.

5.5 Aer terminates the process on deep circuits under Windows

The process dies with `STACK_OVERFLOW` and no Python traceback. The cause is that Aer recurses proportionally to circuit depth while a Windows thread receives a 1 MB stack against Linux’s 8 MB. `run_with_deep_stack()` runs every Aer call on a 64 MB thread, after which the 12-qubit circuit completes in 0.71 s. Note that 256 MB and above fail to allocate the thread on Windows, so the figure is not a knob to raise casually.

Table 6: Failure is a function of circuit depth, not qubit or gate count.

Circuit	Gates	Depth	Result
UCCSD 8q	1,950	1,646	fine
UCCSD 10q	4,969	4,286	fine
UCCSD 12q	12,657	11,127	hard crash, 0xC00000FD
random 12q	12,000	3,139	fine
random 16q	12,000	2,390	fine

5.6 UCCSD removes the penalty apparatus entirely

Because UCCSD conserves particle number exactly — measured leak 2.8×10^{-16} at the optimum, with no penalty term present — there is no penalty to tune, no sector to validate against a penalised operator, no contamination budget doing work, and no SYMMETRY BROKEN state reachable. The hardware-efficient path needs all of it, and on the verification system it leaves 256 mHa of error where UCCSD leaves 0.978 mHa.

The guarantee is narrower than it is sometimes taken to be: UCCSD conserves N and S_z , but **not** S^2 at arbitrary amplitudes, since its $\alpha \rightarrow \alpha$ and $\beta \rightarrow \beta$ amplitudes are independent. At random angles the verification system measures $\langle \hat{S}^2 \rangle = 1.406$. It becomes a spin eigenstate at the variational minimum — $\langle \hat{S}^2 \rangle = 1.12 \times 10^{-4}$ there — which is where the check is applied.

6 The chemistry that has not been run

No molecular results are reported

No Hamiltonian has been built for either the `ts` or the `far` geometry. This report contains no energy for acrylamide plus methanethiolate, no TS–FAR difference, and no comparison against a classical ladder for this system. Every number above is either from the verification system or from a timing measurement.

6.1 Why

Building the active-space Hamiltonian requires PySCF, for the restricted Hartree–Fock, the C-PCM solvation model, the MP2 natural orbital construction and the CASCI reference. PySCF publishes **no Windows wheel**; the only distribution for this platform is a source tarball, and its build is unsupported on Windows even with a toolchain present. This was confirmed on the machine used for this report: `pip` resolves `pyscf-2.14.0.tar.gz` and no binary distribution, and no WSL distribution is installed to run it under.

The workflow was designed for this constraint rather than blocked by it. The expensive chemistry is a one-time step whose output is a JSON cache carrying a specification hash; every downstream command reads the cache and runs natively on Windows. The split is exactly two commands:

Command	Requires	Runs on
<code>prepare, ladder</code>	PySCF	Colab / WSL / Linux / macOS
<code>validate</code>	OpenFermion, SciPy	anywhere
<code>selftest, backend, vqe</code>	Qiskit, Aer	Windows, natively

6.2 What remains

1. Run `colab_prepare.py -active-spaces 8q -geometries ts,far` on a free Colab session. This performs both RHF calculations with C-PCM, builds MP2 natural orbitals, selects the active

space, runs the classical ladder, and packages the caches and receipts for download. The dominant cost is CCSD(T) on aug-cc-pVDZ on approximately two vCPUs; `-skip-ladder` defers it.

2. Validate and run, on any machine: `run.py validate` with `-cache hamiltonian_ts_8q.json -number-penalty 0`, then `run.py vqe` on each of the two caches. At eight qubits with UCCSD the projected cost is 0.6 hours per geometry (Table 4).
3. Report the TS–FAR difference beside the CASCI–CCSD(T) gap from the ladder. The latter measures how far the chosen active space sits from converged chemistry, and is almost always much larger than the VQE’s error against CASCI — so quoting the VQE error alone would overstate what the calculation establishes.

6.3 One anticipated obstacle

A vacuum anion in a fully augmented basis frequently has a *positive* HOMO, in which case the frontier orbitals describe the diffuse tail of the wavefunction rather than the C–S reaction region. The active-space guard in `prepare` detects this and refuses to build rather than producing a Hamiltonian for the wrong physics. MP2 natural orbitals usually resolve it, which is a second reason the published protocol uses them. If the guard still fires, the remedies in order are: restrict diffuse functions to the reacting atoms via a per-element basis dictionary; or override with `-allow-delocalized-active-space` — and in that case the result must not be reported as describing the reaction region.

7 Claims not supported by these data

1. **Any statement about the thio-Michael reaction.** No molecular calculation was performed.
2. **That the ansatz ordering in Table 3 transfers to the real system.** The verification system is random. Its correlation energy, 374.58 mHa, is roughly an order of magnitude larger than a CAS(4e,4o) molecular active space would typically carry, so it is a harder test in absolute terms but not a representative one. The ordering is a hypothesis for the real system, not a measurement of it.
3. **That 16 qubits is a universal ceiling.** It is the ceiling for this workflow, these two ansatz families, this machine (8 logical cores), and a 50-iteration convergence assumption. A different optimiser, a shallower ansatz, or a GPU changes it.
4. **That the TS–FAR difference would be a reaction barrier.** The two geometries are *rigid*. Their difference is a rigid-scan energy difference along an approach coordinate, not a relaxed barrier.
5. **Anything about hardware feasibility.** Every measurement here is a noiseless simulation.

8 Limitations

1. The verification system is a single random instance at a single seed. The checks are exact, so this is adequate for correctness, but the ansatz comparison of §4 is a single sample.
2. Timings in Tables 4 and 5 beyond 14 qubits are projections from measured per-gradient costs, not completed runs.
3. The Aer stack finding is specific to Windows and to the Aer version pinned in `requirements.txt`.

4. Chemical accuracy, once molecular results exist, will mean agreement with CASCI for the cached Hamiltonian — a statement about the solver, not about the basis set or the active space.

9 Conclusions

1. The solver is verified. Twenty checks against exact classical answers pass, including a determinant-by-determinant test of spin-orbital ordering that no spectral check could replace.
2. On the verification system UCCSD recovers 99.74% of the correlation energy at 0.978 mHa above exact FCI in 8.7 seconds, with structurally conserved particle number and no penalty term.
3. k -UpCCGSD at $k \geq 2$ is more accurate and disproportionately more expensive; ADAPT-VQE reproduced UCCSD’s operator set exactly and is not selected automatically at this size.
4. Five implementation findings are documented, each of which would otherwise have produced a wrong answer, a crashed process, or a $30\times$ slower one.
5. No chemistry has been run, for a specific and stated reason, and §6 gives the three commands and the projected cost that would close the gap.

A Software environment

Two environments are recorded. The measurement tables of §4–§5 were produced on the first; the verification run reproduced for this report was executed on the second, and gives 20/20 on both.

	Reference environment	Reproduction environment
Python	3.12.13	3.13.7
qiskit	2.5.2	2.3.1
qiskit-aer	0.17.2	0.17.2
qiskit-nature	0.8.0	0.7.2
OpenFermion	1.8.1	1.8.1
NumPy	2.5.2	2.4.3
SciPy	1.18.0	1.17.1
PySCF	absent	absent
OS	Windows 11	Windows 11

qiskit-nature 0.8.0 predates Qiskit 2.x, so the pairing was tested rather than assumed: UCCSD builds, simulates on Aer, and places $\theta = 0$ exactly on the Hartree–Fock determinant.

B Reproduction

```
uv venv --python 3.12 .venv && .venv\Scripts\activate
uv pip install -r requirements.txt
```

```
python run.py selftest
python run.py backend
python -m unittest discover -s tests -v
```

Then, on a machine with PySCF:

```
pip install -q pyscf openfermion "qiskit==2.5.*" \
    "qiskit-aer==0.17.*" "qiskit-nature==0.8.*"
python colab_prepare.py --active-spaces 8q --geometries ts,far
```

Unzip the resulting archive into the workflow folder, then anywhere:

```
python run.py validate --cache hamiltonian_ts_8q.json --number-penalty 0
python run.py vqe      --cache hamiltonian_ts_8q.json
python run.py vqe      --cache hamiltonian_far_8q.json
python run.py ladder
```

C Data inventory

Artifact	Status
report/figures/acr-ansatz.png	present, from Table 3
report/figures/acr-cost.png	present, from Table 3
report/figures/acr-scaling.png	present, from Tables 4–5
acrylamide_thiolate_hamiltonian.json	not built
acrylamide_thiolate_vqe_result.json	not built
acrylamide_thiolate_classical_ladder.json	not built

The three absent files are the entire chemistry output of this workflow. Their absence is the subject of §6 and is the reason this document is titled a verification report.

References

1. CovAngelo, *Covalent inhibitor reaction energetics on quantum hardware*, arXiv:2604.10487 — the simulator protocol followed here, and the source of Table 1.
2. J. Lee *et al.*, *Generalized unitary coupled cluster wavefunctions for quantum computation*, J. Chem. Theory Comput. **15**, 311 (2019) — *k*-UpCCGSD.
3. H. R. Grimsley *et al.*, *An adaptive variational algorithm for exact molecular simulations on a quantum computer*, Nat. Commun. **10**, 3007 (2019) — ADAPT-VQE.